

Improving the Performance of Inkjet Printed QLED by Annealing Post-treatment

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Abstract

The quantum dot light-emitting diodes (QLEDs) by inkjet printing are promising candidates for lighting and display applications of the industry. Herein, we report an inkjet printed QLED, showing that the luminance of red QLED is as high as 185, 607 cd/m², the external quantum efficiency reaches 12.6 %, and the lifetime (T₉₅) is 549 h @ 1000 cd/m², which is one of the highest reported for a very primitive device structure.

Author Keywords

Inkjet printing, annealing, QLED

1. Introduction

Colloidal quantum dots light-emitting diodes (QLED) have achieved great attention due to their outstanding property, such as size-controlled emission wavelengths, narrow emission bandwidths, and solution-processability(1–7). Since the first report for QLED is in the 1990s, the EL and efficiency have a great progress during past 20 years(8). Up to date, the champion device is comparable to the organic LEDs(9). However, the QLED with high luminance and efficiency were mainly fabricated by spin coating. The process is not amicable to mass production, moreover, spin coating results in loss of more than 94% of QDs solution(10). Meanwhile, spin coating, it is still difficult to scale up, which limits the business application of QLED in the display field. Inkjet printing is particularly attractive for its advantages like no-contact process, on-demand jetting, mask-free, and flexible substrate capability.

However, the low efficiency and stability of IJP-QLED hinder its development and application. This is mainly due to Auger recombination (AR), Foster resonance energy transfer (FRET) and quenching of QD(11). In 2009, Jabbour et al. first report the fabrication of QLED by inkjet printing. However, the EL and efficiency are unsatisfactory. Therefore, it is still very crucial to develop a novel QD ink and corresponding processing methodology to obtain uniform QD layer for QLED. Recently, ink engineering and post-treatment are reported to improve the quality of layer, which can improve the efficiency and stability of inkjet printed QLED. Besides, the limitation of IJP technology on quantum dot solvents (as quantum dot inks) also causes a loss of efficiency and stability (12–15). Therefore, many strategies have been carried out to solve these problems, including adjusting the structure and ligands of the internal quantum

dots(16), developing suitable ink systems(13,17,18), and new post-processing(19). However, there no reports show the device post-treatment to improve the inkjet printed QLED. Hence, we develop an annealing method to improve the efficiency and stability of inkjet printed QLED.

Herein, we optimized the red IJP QLED by heating treatment. The device demonstrated the max EQE of 12.6 %, which be increased productivity threefold compared with control device.

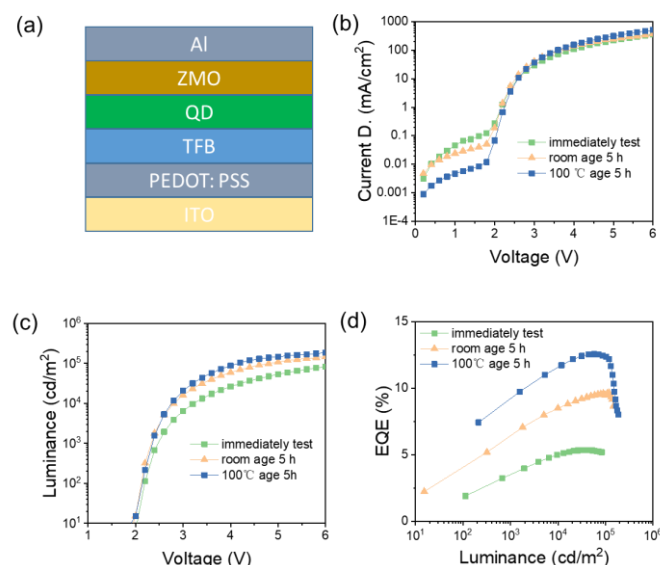


Fig. 1 The performance of inkjet printed QLED under different post-treatment. (a) the structure of inkjet printed QLED. (b) J-V (c) J-L, and (d) EQE-L of the devices.

2. Result and discussion

Fig. 1(a) shows the structure of the inkjet printed QLED. Fig.1(b) shows the J-V of the devices by different post-treatment. after annealing, the leakage current decrease significantly. Meanwhile, the luminance increases up to 185, 607 cd/m². The EQE of QLED by annealing shows 12.6 %, which indicates the annealing process help to accelerate the positive aging. Besides, the device with annealing shows a higher current density and

luminance, which indicate the effective carrier recombination.

Fig. 2(a) shows the operating lifetime of the inkjet printed QLED. The luminance of red QLED is as high as 185,607 cd/m^2 , the external quantum efficiency reaches 12.6 %, and the lifetime (T_{95}) is 549 h @ 1000 cd/m^2 , which is one of the highest reported for a very primitive device structure. Besides, the spectrum has no shift after ageing, indicating that the photoproperties of the QD layer were stable after treatment.

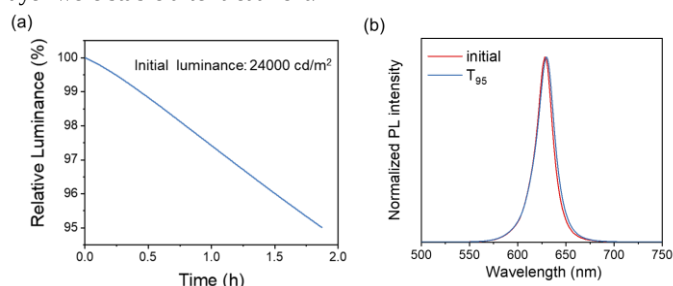


Fig. 2 (a) the lifetime of QLED by annealing post-treatment. (b) the electroluminescence spectrum of the device at initial and T_{95} .

3. Conclusion

Based this method, the luminance of red QLED is as high as 185,607 cd/m^2 , the external quantum efficiency reaches 12.6 %, and the lifetime (T_{95}) is 549 h @ 1000 cd/m^2 , which is one of the highest reported for a very primitive device structure. The annealing post-treatment can help to improve the efficiency and stability of inkjet printed QLED. We investigated the annealing process can accelerate the positive aging of QLED.

4. Experimental

Materials. For red QD, cadmium acetate, zinc acetate, oleic acid, and 1-octadecene were mixed and evacuated under 120 °C for 1 hour and filled with Ar atmosphere. The Se precursor was injected at 260 °C and reacting for 30 min. Zinc oleate and S precursor were injected at 280 °C for 1 h. The vessel was allowed to cool down to room temperature slowly. Then, the solution was purified with ethanol and n-hexane three times. Finally, the QD solution was dispersed in mixed solvent for preparing the ink.

Device fabrication. All ITO glasses were cleaned in acetone, DI-water, and isopropanol for respectively 30 min by using an ultrasonic cleaner. Then, these were treated in a UV-Ozone chamber for 10 min. A PEDOT: PSS layer was deposited on it and annealed at 130 °C for 20 min. Then, a TFB layer was deposited and annealed at 100 °C for 10 min. After that, a QD layer was sequentially printed in the glovebox, and thicknesses of the QD layer was printed by using a Microfab JETLAB 2 printer equipped with a piezoelectric-driven inkjet nozzle (diameter: 40 μm) in glovebox. The post-treatments were applied using a molecular pump. After the deposition of ZnMgO nanoparticles, the substrate was annealed at 80 °C for 20 min. The top electrodes were deposited by heat evaporation of aluminum for 100 nm, followed by packaging using UV-glue in the glove box.

Characterizations. The J-V-L and EQE characteristics of the devices were measured using a QLED characterization system XPQY-EQE-350-1100 (Guangzhou Xi Pu Optoelectronics Technology Co., Ltd.) The lifetime test was conducted using a lifetime test system with photodiode holders from Guangzhou Cryscos Equipment Co., Ltd. All characterization is performed under ambient conditions.

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Reference

1. Dunlap-Shohl WA, Zhou Y, Padture NP, Mitzi DB. Synthetic Approaches for Halide Perovskite Thin Films. *Chem Rev.* 2019;119(5):3193–295.
2. Yang J, Hahm D, Kim K, Rhee S, Lee M, Kim S, et al. High-resolution patterning of colloidal quantum dots via non-destructive, light-driven ligand crosslinking. *Nat Commun [Internet].* 2020;11(1):1–9. Available from: <http://dx.doi.org/10.1038/s41467-020-16652-4>
3. Cho H, Kim YH, Wolf C, Lee HD, Lee TW. Improving the Stability of Metal Halide Perovskite Materials and Light-Emitting Diodes. *Adv Mater.* 2018;30(42):1–24.
4. Chen Y, Xu Z, Smith C, Sankar J. Recent advances on the development of magnesium alloys for biodegradable implants. *Acta Biomater [Internet].* 2014;10(11):4561–73. Available from: <http://dx.doi.org/10.1016/j.actbio.2014.07.005>
5. De San Luis A, Bonnefond A, Barrado M, Guraya T, Iturrondobeitia M, Okariz A, et al. Toward the minimization of fluorescence loss in hybrid cross-linked core-shell PS/QD/PMMA nanoparticles: Effect of the shell thickness. *Chem Eng J [Internet].* 2017;313:261–9. Available from: <http://dx.doi.org/10.1016/j.cej.2016.12.048>
6. Chen S, Cao W, Liu T, Tsang SW, Yang Y, Yan X, et al. On the degradation mechanisms of quantum-dot light-emitting diodes. *Nat Commun [Internet].* 2019;10(1):1–9. Available from: <http://dx.doi.org/10.1038/s41467-019-08749-2>
7. Koppens FHL, Mueller T, Avouris P, Ferrari AC, Vitiello MS, Polini M. Photodetectors based on graphene, other two-dimensional materials and hybrid systems. *Nat Nanotechnol [Internet].* 2014;9(10):780–93. Available from: <http://dx.doi.org/10.1038/nnano.2014.215>
8. Loss D, DiVincenzo DP. Quantum computation with quantum dots. *Phys Rev A - At Mol Opt Phys.* 1998;57(1):120–6.
9. Pu C, Dai X, Shu Y, Zhu M, Deng Y, Jin Y, et al. Electrochemically-stable ligands bridge the photoluminescence-electroluminescence gap of quantum dots. *Nat Commun [Internet].* 2020;11(1):1–10. Available from: <http://dx.doi.org/10.1038/s41467-020-14756-5>
10. Derby B. Inkjet Printing of Functional and Structural Materials: Fluid Property Requirements, Feature Stability, and Resolution. *Annu Rev Mater Res.* 2010;40(1):395–414.
11. Cao W, Xiang C, Yang Y, Chen Q, Chen L, Yan X, et al. Highly stable QLEDs with improved hole injection via quantum dot structure tailoring. *Nat Commun [Internet].* 2018;9(1):2–7. Available from: <http://dx.doi.org/10.1038/s41467-018-04986-z>
12. Singh M, Haverinen HM, Dhagat P, Jabbour GE. Inkjet printing-process and its applications. *Adv Mater.* 2010;22(6):673–85.

13. Jiang C, Zhong Z, Liu B, He Z, Zou J, Wang L, et al. Coffee-Ring-Free Quantum Dot Thin Film Using Inkjet Printing from a Mixed-Solvent System on Modified ZnO Transport Layer for Light-Emitting Devices. *ACS Appl Mater Interfaces*. 2016;8(39):26162–8.
14. Lim B, Hwang JT, Jin YK, Ghim J, Vak D, Noh YY, et al. Synthesis of a new cross-linkable perfluorocyclobutane-based hole-transport material. *Org Lett*. 2006;8(21):4703–6.
15. Brennan MC, Zinna J, Kuno M. Existence of a Size-Dependent Stokes Shift in CsPbBr₃ Perovskite Nanocrystals. *ACS Energy Lett*. 2017;2(7):1487–8.
16. Xiang C, Wu L, Lu Z, Li M, Wen Y, Yang Y, et al. High efficiency and stability of ink-jet printed quantum dot light emitting diodes. *Nat Commun* [Internet]. 2020 Dec 2;11(1):1646. Available from: <http://dx.doi.org/10.1038/s41467-020-15481-9>
17. Tang P, Xie L, Xiong X, Wei C, Zhao W, Chen M, et al. Realizing 22.3% EQE and 7-Fold Lifetime Enhancement in QLEDs via Blending Polymer TFB and Cross-Linkable Small Molecules for a Solvent-Resistant Hole Transport Layer. *ACS Appl Mater Interfaces* [Internet]. 2020 Mar 18;12(11):13087–95. Available from: <https://pubs.acs.org/doi/10.1021/acsami.0c01001>
18. Li Z, Li P, Chen G, Cheng Y, Pi X, Yu X, et al. Ink Engineering of Inkjet Printing Perovskite. *ACS Appl Mater Interfaces*. 2020;12(35):39082–91.
19. Han YJ, Kim DY, An K, Kang KT, Ju BK, Cho KH. Sequential improvement from cosolvents ink formulation to vacuum annealing for ink-jet printed quantum-dot light-emitting diodes. *Materials (Basel)*. 2020;13(21):1–14.